STAT GU4243/5243 Project 1 Report

HOME CREDIT DEFAULT RISK: DATA ACQUISITION, CLEANING, PREPROCESSING, AND FEATURE ENGINEERING

Course: STAT GU4243/5243
Github Repository: https://github.com/olddoorhead/5243Project1
Primary Notebook: `Project_1.ipynb`

EXECUTIVE SUMMARY

- **Data integration:** We consolidated all core `Home Credit Default Risk` relational sources into a joined application-level table of **356,255 rows x 894 columns**.
- **Split discipline:** We used stratified 80/20 train-validation splitting first, preserving class balance at **0.080729** (train) and **0.080728** (validation).
- **Feature control:** Cleaning and pruning reduced feature width from **891** to **396** (**495 net features removed**) while preserving model-ready structure.
- **Validation gains:** Grouped imputation and winsorization improved validation ROC-AUC from **0.693368** (no clipping baseline) to **0.708085**.
- **Imbalance handling:** Using SMOTENC on training set only to raise minority prevalence from **8.07%** to **16.67%** and added **25,369** synthetic training rows.
- **Final artifacts:** Exported matrices are consistent for downstream modeling: Train **(271,377, 396)**, Validation **(61,503, 396)**, Test **(48,744, 396)**.

RUBRIC COVERAGE AND EVIDENCE

Requirements	Evidence location
Data acquisition methodology	Section 1.2, Section 1.3, Figure 1, Figure 2, Appendix Figure A1
Data cleaning and inconsistency handling	Section 1.4, Section 4.1, Figure 8, Appendix Table A1, Appendix Table A2, Appendix Table A6
EDA with patterns, distributions, anomalies	Section 3, Figure 5-Figure 8, Appendix Figure A2-Appendix Figure A7
Preprocessing and feature engineering with justification	Section 2.2, Section 4.1, Table 2, Table 4, Figure 4, Figure 9, Appendix Table A4, Appendix Table A5, Appendix Table A7
Structured methodology and findings	Sections 1-8, Figure 3, Table 2, Table 3, Table 4
Summary of key findings	Section 6
Challenges and limitations	Section 7
Future recommendations	Section 8
Public GitHub reference	Header repository link
Team member contribution statement	Section 9

Table 1: Rubric-to-report evidence mapping.

Hint: Evidence labels prefixed with `A` (for example, `Appendix Figure A4` and `Appendix Table A5`) are in `Appendix.md`.

1. PROBLEM STATEMENT AND DATA

1.1 Objective

Develop an auditable, leakage-aware preprocessing and feature-engineering pipeline for default-risk modeling readiness on the Home Credit dataset. The project scope explicitly covers acquisition, cleaning, EDA, preprocessing, feature engineering, and export of finalized modeling datasets.

1.2 Data Acquisition

- **Source:** Kaggle Home Credit Default Risk competition data.
- **Acquisition mode:** Public CSV download and local ingestion.
- **Files used:**
  - **Application tables:** `application_train.csv`, `application_test.csv`

- **Bureau tables:** `bureau.csv`, `bureau_balance.csv`
- **Behavioral history:** `previous_application.csv`, `POS_CASH_balance.csv`
- **Card and installment history:** `credit_card_balance.csv`, `installments_payments.csv`

1.3 Dataset Scope and Split Strategy

- **Application union shape: (356,255, 123)** after train/test schema alignment.
- **Post-aggregation shape: (356,255, 894)** after customer-level joins.
- **Modeling matrices after leakage-safe drops (`TARGET`, `SK_ID_CURR`, `_is_train`):**
  - **Train candidates: (307,511, 891)**
  - **Test candidates: (48,744, 891)**
- **Stratified split on train rows only:**
  - **Training fold: `X_tr_raw` (246,008, 891)**, positive rate **0.080729**
  - **Validation fold: `X_val_raw` (61,503, 891)**, positive rate **0.080728**

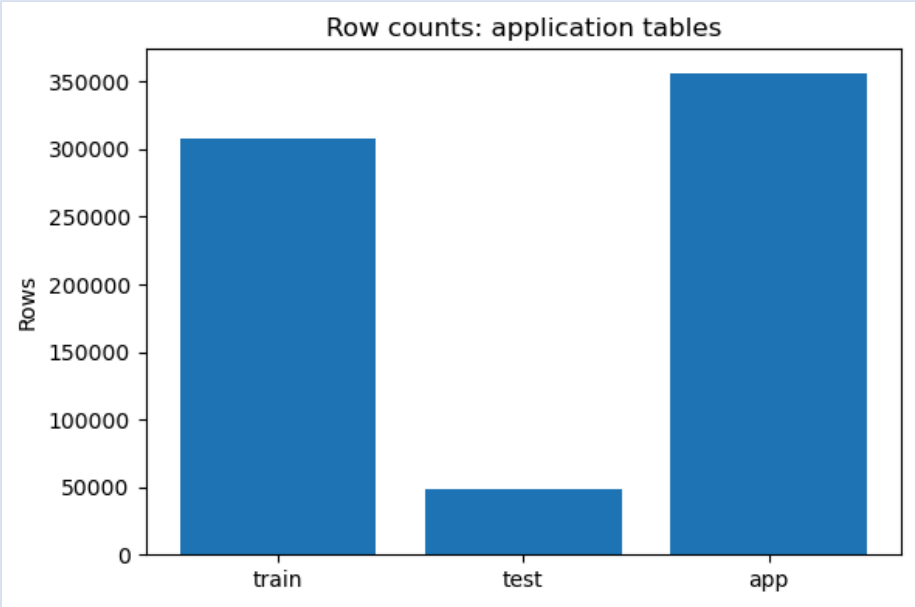

Figure 1: Row counts for application train, test, and combined table.

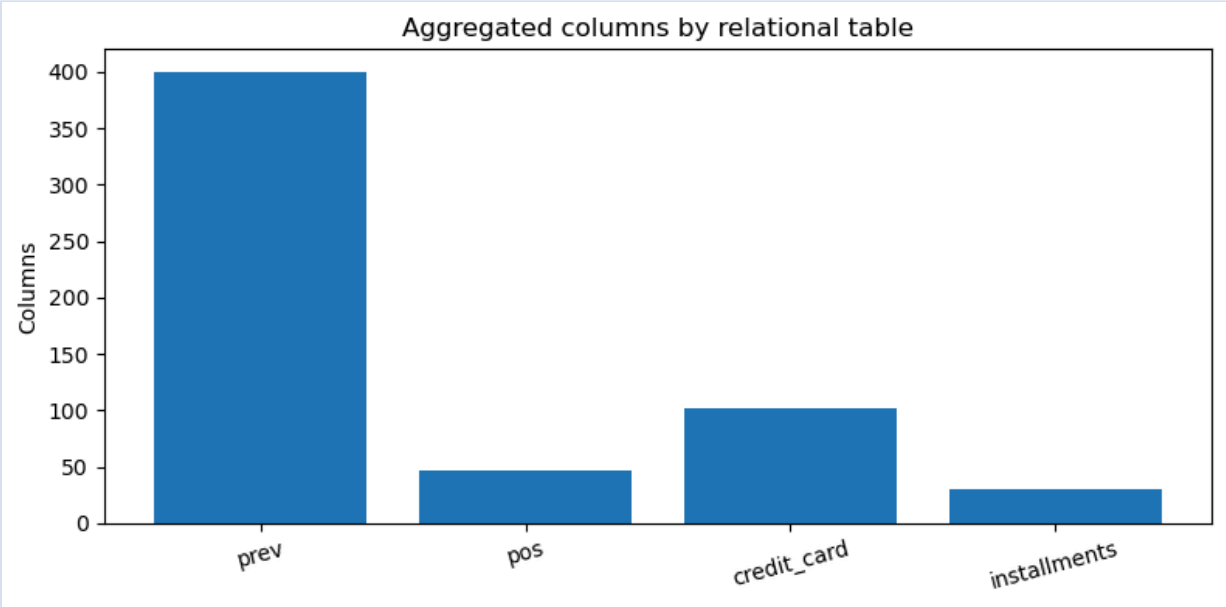

Figure 2: Aggregated column contributions by relational source table.

1.4 Leakage and Reproducibility Controls

- **Split-first protocol:** All fit-dependent transformations were trained on the training fold before validation/test application.
- **Consistent transforms:** Threshold-based feature removal was applied identically across train/validation/test once selected.
- **Traceability:** Feature-drop logs and final matrix exports were written deterministically to `log/` and `cleaned_dataset/`.

2. METHOD AND DESIGN RATIONALE

2.1 Pipeline Architecture

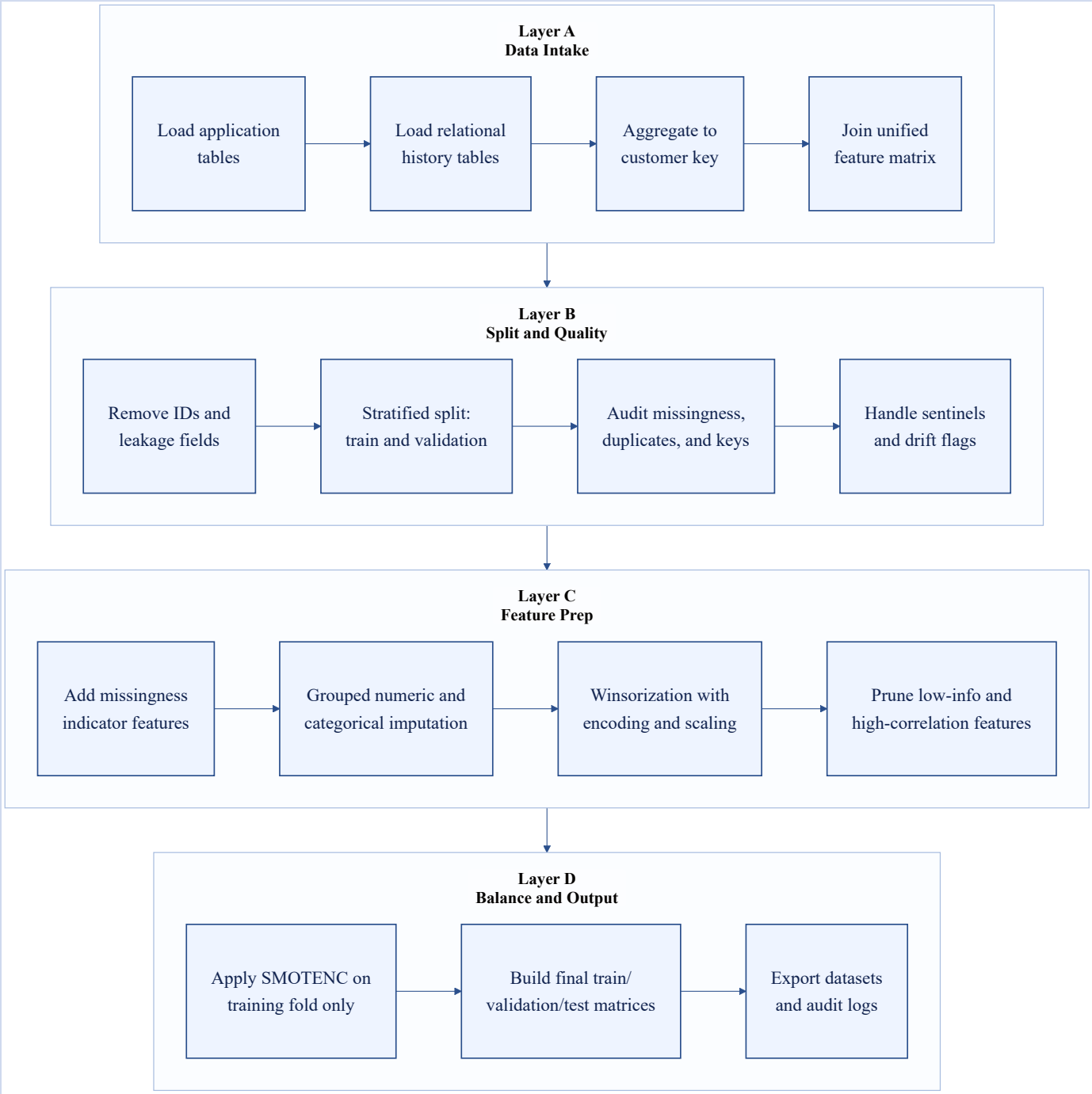

Figure 3: Project Pipeline from Acquisition to Export.

2.2 Design Rationale

The pipeline was intentionally designed around risk-control decisions rather than only maximizing short-run validation scores. The key design choices and supporting evidence are summarized below.

2.2.1 Split-Before-Transform Logic

- **Rationale:** Any transformation learned on full data can leak distributional information from validation/test into training.
- **Implementation:** Stratified split is executed before grouped statistics, scaling parameters, clipping bounds, and feature filtering.
- **Benefit:** Ensures validation metrics reflect realistic out-of-sample behavior and guards against optimistic bias.

2.2.2 PSI Threshold Selection (0.15)

- **Rationale:** PSI values above **0.15** are commonly treated as meaningful distribution drift for monitoring and feature stability review.
- **Implementation:** Numeric features were flagged when **PSI > 0.15**; categorical features were flagged when **L1 share shift > 0.15**.
- **Observed effect:** Three unstable features were removed before final modeling.

2.2.3 Grouped Imputation Choice

- **Rationale:** Global imputation can dilute subgroup structure in credit-risk data.
- **Implementation:** We benchmarked grouped mean/median imputations across candidate grouping fields.
- **Selected policy:** `OCCUPATION_TYPE + Median` achieved the highest validation AUC (**0.701886**), outperforming global median baseline (**0.696289**).

2.2.4 Winsorization Choice

- **Rationale:** Extreme tails in credit variables can destabilize linear decision boundaries and scaling.
- **Implementation:** Quantile clipping bounds were tuned against validation ROC-AUC on a high-variance feature subset.
- **Selected policy: Q0.05-Q0.95** clipping improved AUC to **0.708085** versus **0.693368** without clipping.

2.2.5 Low-Information Pruning

- **Rationale:** Features that are constant or nearly constant across training rows add little predictive value and can introduce avoidable noise.
- **Implementation:** Dropped near-constant features when dominant share `>= 0.999`.
- **Observed effect:** Removed **259 low-information features**, reducing the matrix to **438 features** before correlation pruning.

2.2.6 Correlation Pruning Threshold

- **Rationale:** Highly collinear features increase redundancy and can inflate model variance without adding signal.
- **Implementation:** For feature pairs with correlation above **0.98**, we kept the stronger target-associated feature (or higher variance on ties).
- **Observed effect: 42** highly collinear features were removed, contributing to a compact final matrix.

Design control	Selected value	Practical purpose	Outcome
Drift control	PSI > 0.15; L1 shift > 0.15	Remove unstable features	3 features removed
Grouped imputation	`OCCUPATION_TYPE + Median`	Preserve subgroup signal	+0.005597 AUC vs global median
Winsorization	Q0.05 to Q0.95	Reduce outlier leverage	+0.014717 AUC vs no clipping
Low-Information pruning	Near constant	Remove redundant predictors	259 features removed
Correlation pruning	> 0.98	Remove redundant predictors	42 features removed

Table 2: Design-control decisions and observed outcomes.

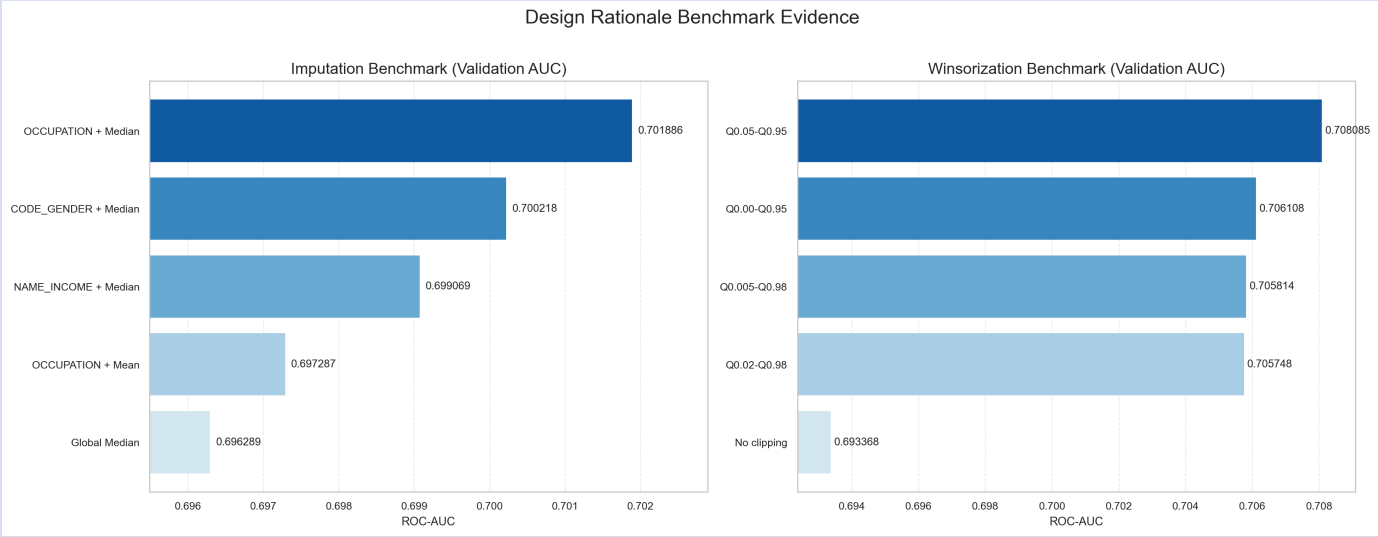

Figure 4: Benchmark evidence supporting grouped imputation and winsorization decisions.

2.3 Alternative Designs Considered and Rejected

The final method was selected after comparing practical alternatives that were technically valid but less suitable for this dataset and assignment scope.

- **Transform-before-split pipeline**
  - **Alternative:** Fit imputers/scalers on the full joined training table before creating holdout folds.
  - **Why rejected:** This introduces subtle information bleed from validation rows into train-fitted preprocessing states.
  - **Expected impact if used:** Inflated validation scores and weaker reproducibility when retraining on fresh data.
- **Stricter drift threshold (PSI > 0.10)**
  - **Alternative:** Remove all features with mild-to-moderate shift.
  - **Why rejected:** At this feature scale, a 0.10 threshold can over-remove useful predictors and reduce signal diversity.
  - **Expected impact if used:** Lower model capacity due to aggressive feature elimination.
- **Looser drift threshold (PSI > 0.25)**
  - **Alternative:** Keep most features unless drift is severe.
  - **Why rejected:** This allows unstable predictors to persist and increases deployment fragility.
  - **Expected impact if used:** Higher risk of train-to-test instability and inconsistent score behavior.
- **Global-only imputation**
  - **Alternative:** Use one median/mean value per feature for all borrowers.
  - **Why rejected:** It collapses subgroup structure (income/occupation profile effects) and reduced validation AUC.
  - **Expected impact if used:** Weaker discrimination in heterogeneous borrower cohorts.
- **More aggressive clipping (e.g., Q0.01-Q0.99 or one-sided heavy clipping)**
  - **Alternative:** Clip fewer or more extreme values depending on tail assumptions.
  - **Why rejected:** Benchmarks showed inferior AUC compared with Q0.05-Q0.95.
  - **Expected impact if used:** Either residual outlier leverage (too weak) or excessive signal compression (too strong).
- **Lower correlation threshold (0.95)**
  - **Alternative:** Remove broader groups of correlated features.
  - **Why rejected:** It risks dropping too many informative near-duplicate aggregates in relational credit features.
  - **Expected impact if used:** Reduced feature richness without clear robustness gain.
- **Higher correlation threshold (0.995)**
  - **Alternative:** Prune only near-identical columns.
  - **Why rejected:** Leaves too much redundancy and raises downstream variance/multicollinearity risk.
  - **Expected impact if used:** Less stable coefficient-based models and noisier interpretation.

2.4 Design Risk Register

Design decision	Main risk if mis-specified	Mitigation used in this project
Split-before-transform	Holdout leakage through preprocessing statistics	Train-fold fitting only; strict fold isolation
Drift filtering thresholds	Over-pruning or under-pruning unstable features	Balanced thresholds with explicit audit counts
Grouped imputation policy	Overfitting to sparse groups or weak global fallback	Group + global fallback + validation benchmarking
Winsorization bounds	Signal loss from over-clipping or outlier dominance from under-clipping	Quantile grid search using validation ROC-AUC
Correlation pruning cutoff	Loss of useful predictors or retained redundancy	Signal-aware pairwise selection and variance tie-breaks

Table 3: Design-risk register and mitigation actions.

3. EXPLORATORY DATA ANALYSIS

3.1 Target Imbalance

- **Class profile:** Training labels are heavily skewed (about **91.9%** class 0, **8.1%** class 1).
- **Implication:** Imbalance-aware preprocessing is required before classifier training.

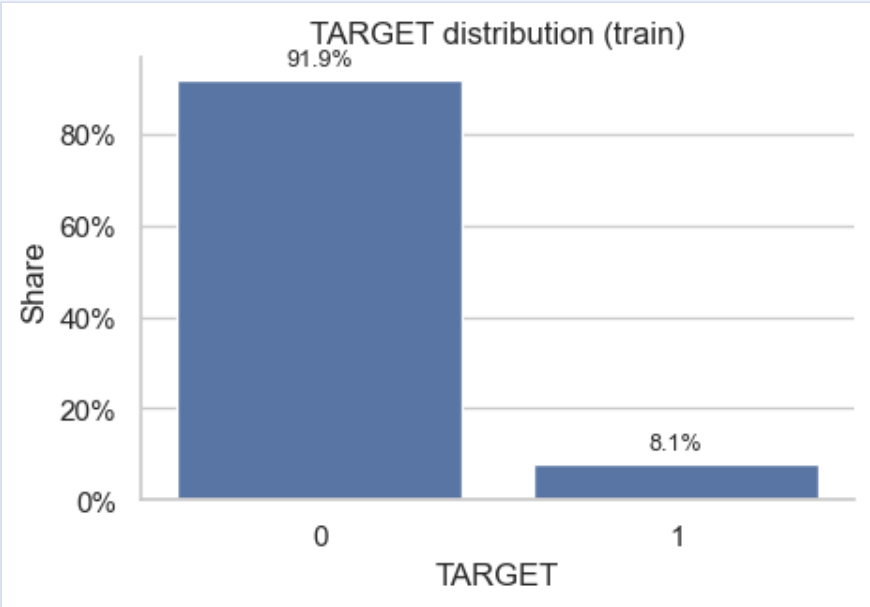

Figure 5: Target-class distribution in the original training data.

3.2 Missingness Patterns

- **Broad profile: 75.8%** of features have missing rate below 0.5.
- **High-missing block: 24.2%** of features (**216 columns**) have missing rate at or above 0.5.
- **Structural signal:** Missingness is concentrated in specific relational families rather than uniformly random.

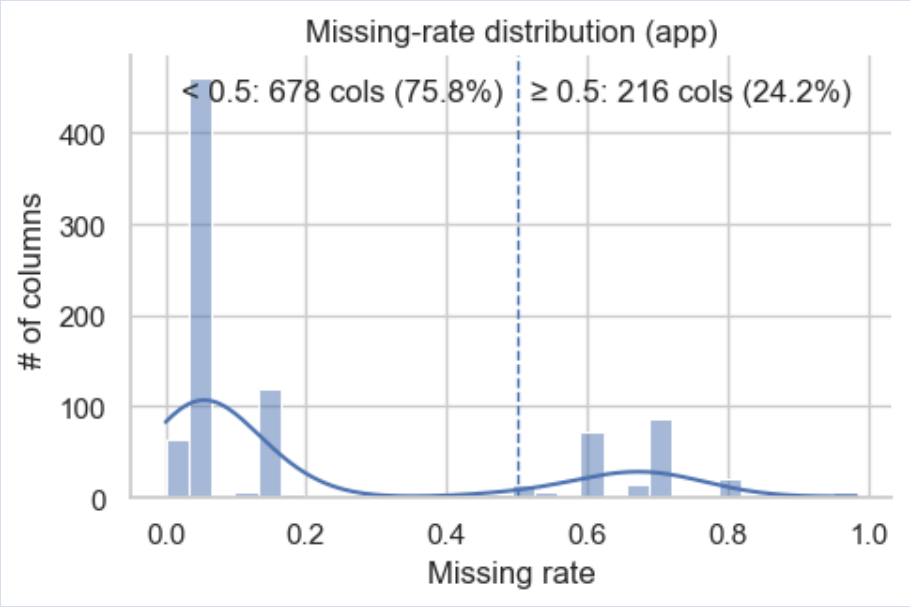

Figure 6: Distribution of missing rates across joined application-level features.

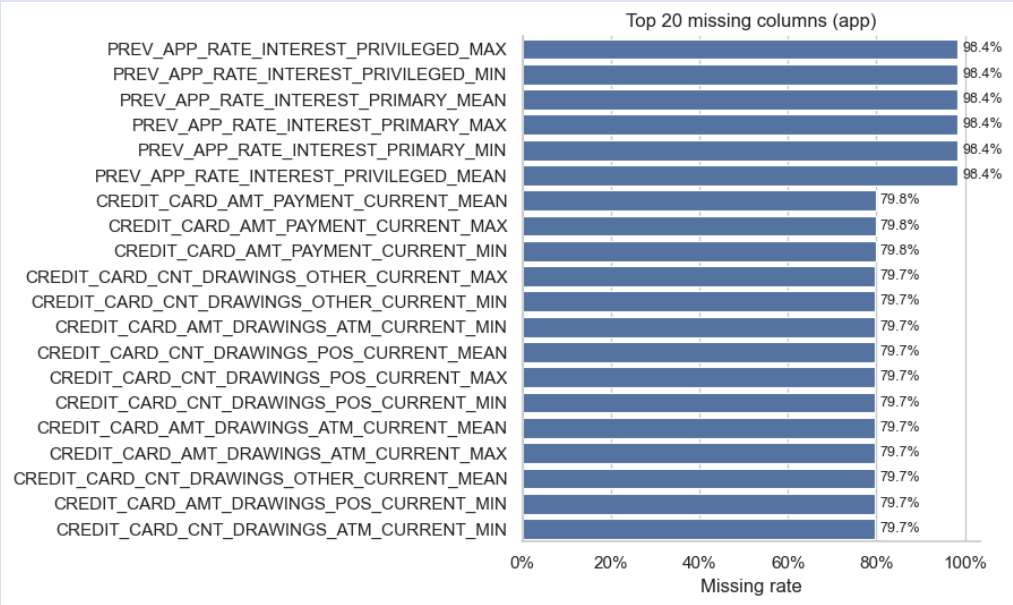

Figure 7: Highest-missing columns in the joined feature space.

3.3 Sentinel and Integrity Checks

- **Sentinel diagnostic:** `DAYS_EMPLOYED = 365243` appears in roughly **18%** of rows and was explicitly handled.
- **Key integrity:** Duplicate-row and primary-key audits showed no duplicate keys in major joined tables.

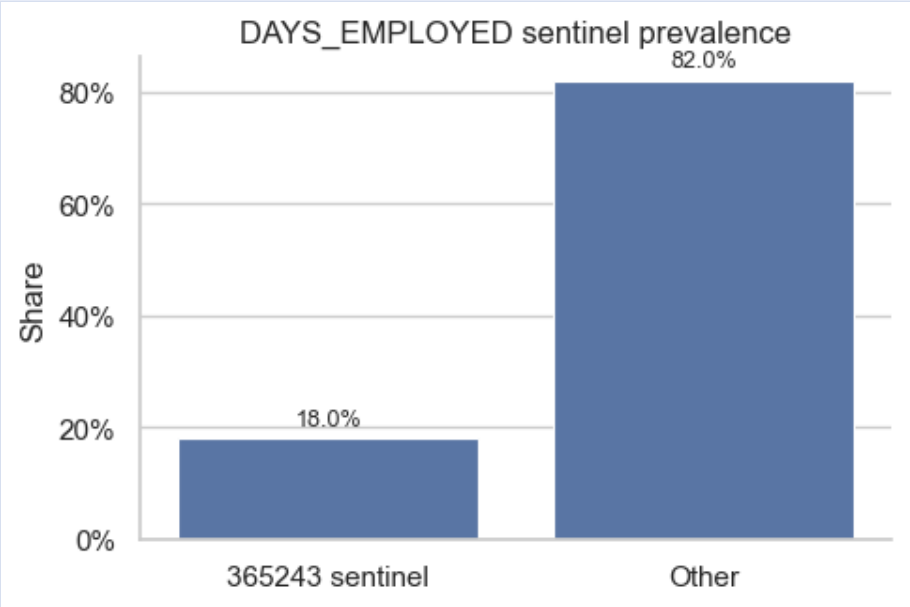

Figure 8: Prevalence of sentinel DAYS_EMPLOYED values.

4. PREPROCESSING AND FEATURE ENGINEERING

4.1 Core Processing Sequence

- **High-missing filter:** Threshold **0.65** on training fold, removing **196** features.
- **Drift filter:** PSI/L1 thresholds from Section 2.2 removed **3** unstable features.
- **Indicator features:** Added social-circle, bureau-request, and phone-change missingness flags.
- **Numeric imputation:** Applied selected grouped median policy using train-fitted statistics.
- **Categorical imputation:** Filled categorical NaNs with explicit `Missing` token.
- **Outlier handling:** Applied selected winsorization bounds to continuous numeric features.
- **Encoding/scaling:** Ordinal encoding plus scaled continuous numerics.
- **Final pruning:** Removed **259** low-information and **42** high-correlation features.

4.2 Feature-Count Audit

Stage	Training shape	Change vs prior
Pre-pruning matrix	(246,008, 891)	-
After high-missing and drift filtering	(246,008, 694)	-197
After indicator additions	(246,008, 698)	+4
Final selected matrix	(246,008, 396)	-302

Table 4: Feature-count changes across the preprocessing pipeline.

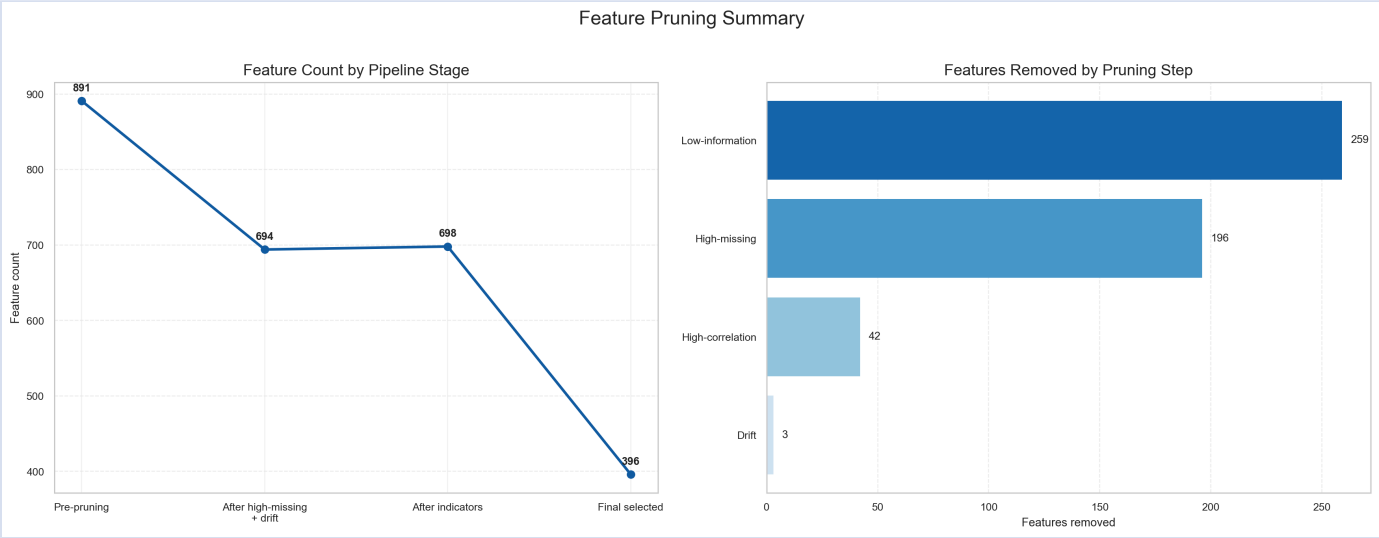

Figure 9: Side-by-side pruning summary (stage counts and per-step removals).

5. MODELING-ORIENTED VALIDATION RESULTS

5.1 Metric and Evaluation Setup

- **Validation protocol:** Stratified holdout split from original training rows.
- **Primary metric:** ROC-AUC to compare preprocessing strategy variants.
- **Usage:** Results guide preprocessing design, not final classifier ranking.

5.2 AUC Progression

- **No clipping baseline: 0.693368**
- **Global median baseline (grouping disabled): 0.696289**
- **Best grouped imputation: 0.701886**
- **Best winsorized pipeline: 0.708085**

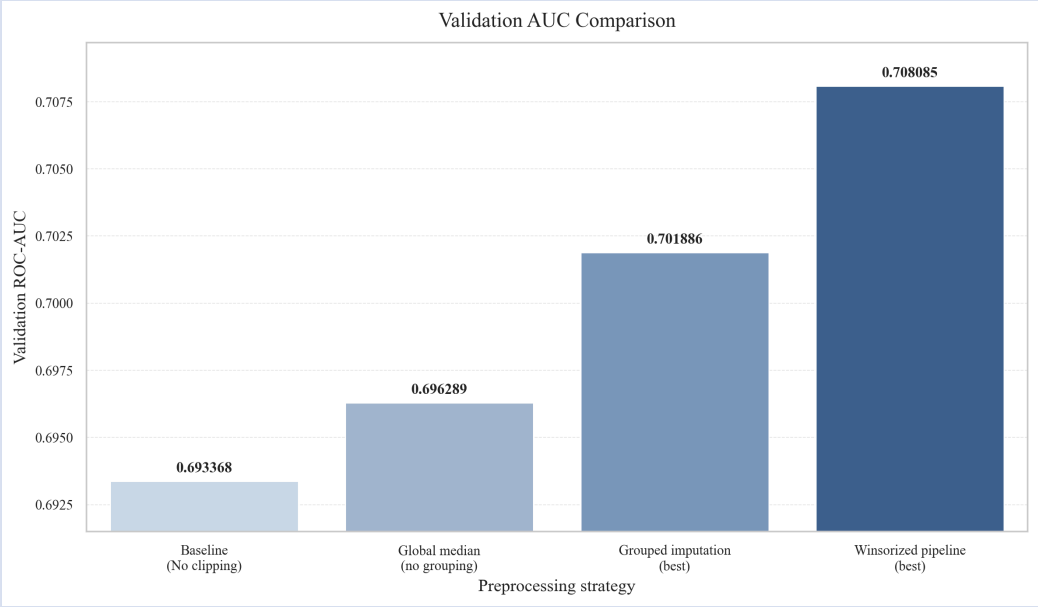

Figure 10: Clean validation AUC comparison of baseline and selected preprocessing configurations.

5.3 SMOTENC Rebalancing Outcome

- **Before SMOTENC:** Training target rate **8.07%**, shape **(246,008, 396)**.
- **After SMOTENC:** Training target rate **16.67%**, shape **(271,377, 396)**.
- **Synthetic increase: 25,369** rows.

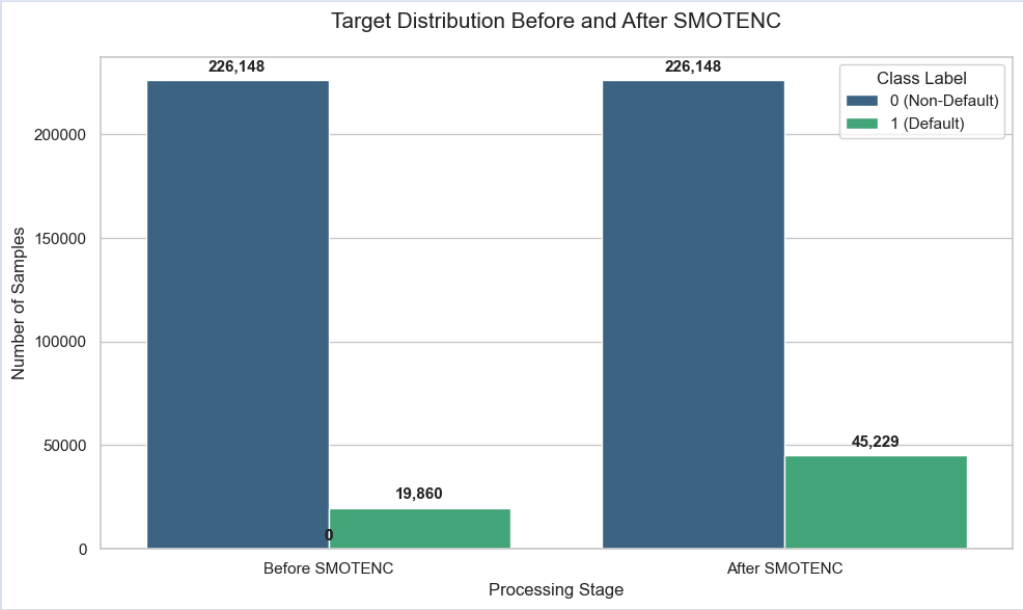

Figure 11: Class-count comparison before and after SMOTENC.

6. KEY FINDINGS

1. **Relational signal consolidation:** Aggregation across bureau, installment, card, and previous-loan tables produced a high-signal candidate feature base.
2. **Risk-controlled preprocessing:** Split-first and threshold-based filtering reduced leakage and instability exposure.
3. **Meaningful validation uplift:** Grouped imputation and winsorization together improved validation discrimination.
4. **Efficient feature space:** The final selected matrix (396 features) preserves key signal while removing substantial redundancy/noise.
5. **Ready outputs:** Final train/validation/test artifacts are aligned for downstream model benchmarking.

7. LIMITATIONS AND CHALLENGES

- **Scope boundary:** This report centers on data pipeline quality; full classifier sweep and tuning remain future work.
- **Drift governance:** Unlabeled test-distribution drift checks are useful for stability but should be controlled carefully in strict blind evaluation pipelines.
- **Interpretability gap:** Current deliverable does not yet include SHAP-level global/local interpretation or subgroup fairness diagnostics.

- **Calibration gap:** Additional probability calibration diagnostics (e.g., reliability curves, Brier score) are still needed.

8. NEXT STEPS

1. **Model benchmarking:** Compare Logistic Regression, LightGBM, and XGBoost on finalized features.
2. **Metric expansion:** Add PR-AUC, recall-at-precision, and calibration diagnostics alongside ROC-AUC.
3. **Ablation testing:** Compare no resampling, SMOTENC, and class-weighted learners.
4. **Monitoring design:** Track drift on top predictors and score distributions with threshold-driven retraining rules.

9. TEAM CONTRIBUTIONS

- **Zhewei Deng:** data integration, EDA diagnostics, missingness analysis, validation checks, report consolidation and packaging of final deliverables.
- **Xiying Chen:** EDA diagnostics, drift/missingness analysis, and quality audit contributions.
- **Kexuan Liu:** feature engineering, imputation/outlier benchmarking, and pruning logic.
- **Yumeng Xu:** report consolidation, result interpretation, and packaging of final deliverables.

There is no significant difference in the effort put in by team members.

Extended tables and additional figures are provided in the separate appendix: `Appendix.pdf`.